

# **xArm 1S**

## **2D Table Pick-and-Place**

Project Report & Engineering Findings

ROS 2 Humble · MoveIt 2 · Custom 5-DOF Numerical IK ·  
Eye-to-hand Astra Pro · ArUco Fiducials · Planar Homography

*Compiled 2026-05-03*

*Workspace: ~/xarm\_moveit · Branch: main*

## Executive Summary

This report documents an end-to-end 2D pick-and-place system built around the **xArm 1S** — a 5-DOF positioning manipulator with a 1-DOF parallel-jaw gripper — observed by a fixed (eye-to-hand) Orbbec Astra Pro RGB-D camera. The objective was to reliably grasp a 40 mm cube (with a 30 mm *DICT\_5X5\_50* ArUco marker, id 2) off a flat table and optionally place it at a target XY on the same plane.

The system has now reached **routinely successful picks**. Two engineering decisions were decisive:

- **Replace 3D hand-eye calibration with a 2D planar homography.** All picks live on a single plane (the table), so a  $3 \times 3$  pixel→world matrix absorbs every linear distortion — camera, table tilt, focal length error — into one SVD-fit object that needs only 6–8 calibration points instead of the dozens of motions required by the  $AX = XB$  hand-eye problem.
- **Tighten the IK joint limits to  $\pm\pi/2$  to match the silent driver clamp.** The hardware driver clamps every commanded joint to  $\pm 90^\circ$  before sending it to the servos. Solutions outside this range were silently truncated, producing ~30 mm of position error and ~17° of unmodelled tilt at the table surface. Constraining IK to the driver's effective range closed the loop.

The IK itself was rewritten in a clean Modern Robotics style (Lynch & Park, 2017) using the Product of Exponentials forward kinematics and a **damped-least-squares Newton-Raphson** with task-priority null-space redundancy resolution: position is the primary task; gripper-down preference is projected into the null space of the position Jacobian so it never fights position. This replaced an earlier scipy L-BFGS-B optimiser. With a brief “polish” phase that continues iterating after position converges to spend the redundant DOFs on orientation, sub-millimetre position error and 5–25° tilts are typical across the reachable workspace.

### Quick numbers

| Metric                      | Value                                 |
|-----------------------------|---------------------------------------|
| Position accuracy (typical) | < 0.1 mm at the IK level              |
| Gripper tilt (typical)      | 6° – 27° from vertical                |
| IK seeds per call           | 24 (multi-start branch coverage)      |
| IK runtime                  | ~1.5 s per solve_ik                   |
| Calibration residual        | ~2.5 mm mean (9/12 inliers)           |
| Workspace (gripper-down)    | ≈ 80 – 200 mm radial ring around base |
| Joint range (IK = driver)   | ±1.5707 rad on all 5 planning joints  |

# Contents

1. Project goal
2. Hardware
3. Software stack
4. End-to-end pipeline
5. Forward kinematics — Product of Exponentials
6. Inverse kinematics — DLS Newton with null-space task priority
7. Calibration — pinhole, IPPE, planar homography
8. Trajectory generation — OMPL RRT-Connect + TOTG
9. Execution & the driver clamp
10. Findings & engineering breakthroughs
11. Test results
12. Known limitations & next steps
13. References & file pointers

## 1. Project goal

Pick a 40 mm cube off a flat table with the xArm 1S and optionally place it at a target XY on the same table. The cube carries a 30 mm ArUco marker (DICT\_5X5\_50, marker\_id=2) on its top face. The camera is mounted on a tripod, fixed in the world (eye-to-hand); the arm and the cube share that world. Long-term goal is multi-cube pick-and-place keyed on per-marker identity; v1 below handles a single cube.

Three constraints shape every choice in this system:

- The arm has **5 actuated DOFs** for positioning. Six-DOF tasks (arbitrary tool orientation) are mathematically underdetermined; the best we can do is achieve a target position and accept the orientation the arm can deliver, then post-filter by tilt.
- Picks happen on a **single planar surface**. This collapses the 3D hand-eye problem to a 2D pixel-to-world homography.
- The hardware driver **silently clamps each joint to  $\pm\pi/2$** . Every component of the IK has to respect this constraint or the predicted FK will not match the executed FK.

## 2. Hardware

| Item    | Detail                                                                                                   |
|---------|----------------------------------------------------------------------------------------------------------|
| Arm     | xArm 1S — 5 DOF positioning + 1 DOF gripper. USB control via the LewanSoul HID protocol; the xarm Python |
| Camera  | Orbbec Astra Pro RGB-D, eye-to-hand on a tripod, fixed in the world frame. Vended ros2_astra_camera dr   |
| Cube    | 40 mm wooden cube. 30 mm ArUco marker (DICT_5X5_50, id 2) printed on its top face.                       |
| Compute | Linux 6.8 / Ubuntu 22.04 host with USB 2.0 access to both arm and camera.                                |

xArm 1S — kinematic chain at home pose

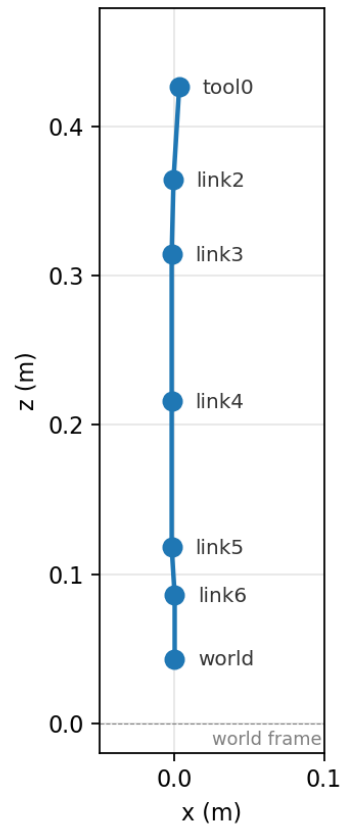


Figure 1. xArm 1S kinematic chain at the home configuration ( $q = 0$ ). The base is at  $z = 0.043$  m; the tool0 frame (gripper midpoint) sits at  $z \approx 0.427$  m at home. The chain is composed by world  $\rightarrow$  base\_link  $\rightarrow$  arm6  $\rightarrow$  arm5  $\rightarrow$  arm4  $\rightarrow$  arm3  $\rightarrow$  arm2  $\rightarrow$  tool0.

### 3. Software stack

| Layer              | Component                                                                          | Notes                                                                                |
|--------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| OS / middleware    | Ubuntu 22.04 + ROS 2 Humble                                                        | —                                                                                    |
| Motion planning    | MoveIt 2 (OMPL / RRT-Connect)                                                      | KDL is bypassed — it returns NO_IK_SOLUTION on this 5-DOF arm.                       |
| Time scaling       | AddTimeOptimalParameterization (Replanned from AddTimeParameterization in Humble). | Replanned                                                                            |
| Hardware bridge    | xarm_hw                                                                            | Bridges /joint_trajectory $\rightarrow$ USB. Implements the silent $\pm\pi/2$ clamp. |
| Vision / fiducials | charuco_tf_publisher (single_aruco_bridge)                                         | Broadcasts camera_color_optical_frame $\rightarrow$ handeye_target.                  |
| Pick stack         | xarm_pick                                                                          | arm_ik.py, calibrate_homography.py, pick_2d.py.                                      |
| Camera driver      | ros2_astra_camera (vendored)                                                       | —                                                                                    |

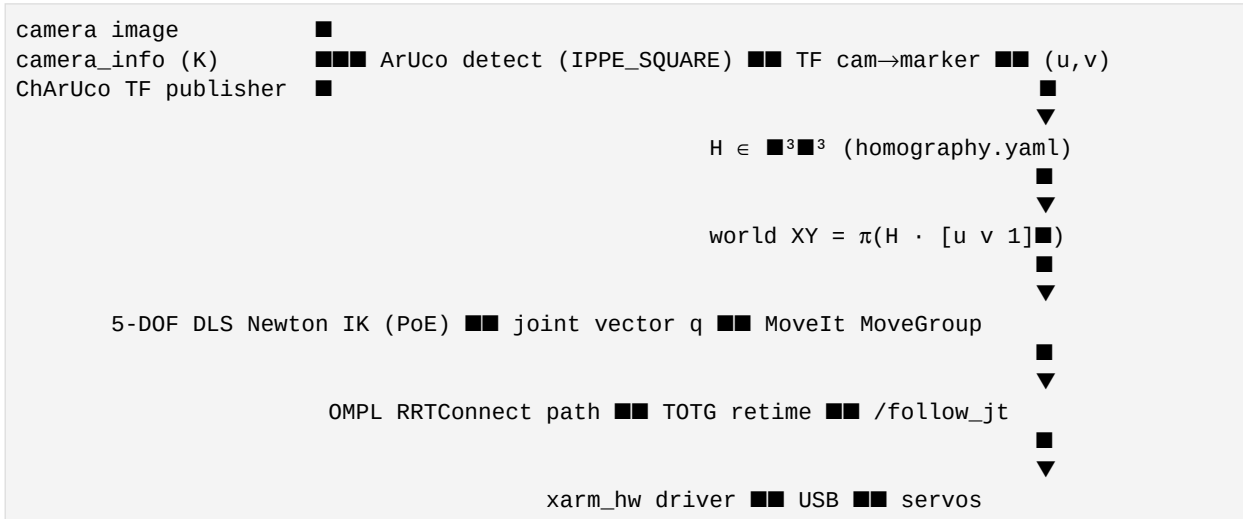
#### 3.1 Bring-up scripts

Five tmux components — *display*, *driver*, *camera*, *marker*, *moveit* — wrap into one bring-up tool with five profiles. Only one bring-up session can run at a time.

| Script              | Components                   | Use case                                  |
|---------------------|------------------------------|-------------------------------------------|
| ./bringup_pick.sh   | all 5                        | Full pick stack (default).                |
| ./bringup_calib.sh  | display + driver + cam + mkr | Run calibrate_homography.                 |
| ./bringup_vision.sh | cam + marker                 | Tune ArUco without arm.                   |
| ./bringup_robot.sh  | display + driver             | Drag-teach, gripper, raw JointTrajectory. |
| ./bringup_moveit.sh | display + driver + moveit    | v1 joint-space, no camera.                |

## 4. End-to-end pipeline

A single pick is the composition of the following stages. Each stage corresponds to one identifiable section in the source tree.



Stages 1–4 happen continuously in vision; stages 5–8 fire once per pick state-machine transition. The driver's 20 Hz JointState feedback closes the loop in tf2\_ros; RViz and pick\_2d see the actual pose.

## 5. Forward kinematics — Product of Exponentials

The forward kinematics is implemented in `arm_ik.py:fk_tool0` using the **Product of Exponentials (PoE)** form (Modern Robotics §4.1):

```
T(θ) = exp([Si]θi) · exp([Si]θi) · ... · exp([Sn]θn) · M

where Si ∈ ℝ6 are screw axes in the SPACE (world) frame at the home pose
and M ∈ SE(3) is Tworld_tool0 at q = 0.

Joint indices: q[0]=arm6 (S1), ..., q[4]=arm2 (S5).
```

Why PoE rather than per-link composition: the kinematics is identical, but the space and body Jacobians (MR §5.1) drop out as analytical functions of the screw axes and the running prefix transform. This makes every Newton iteration analytical — no finite differences.

### 5.1 Screw axes (verified against URDF chain to 0.000 μm)

Each row is  $S_i = (\omega_x, \omega_y, \omega_z, v_x, v_y, v_z)$  in the space frame at home, with  $v = -\omega \times q$  where  $q$  is any point on joint  $i$ 's axis. The arm6 axis is (0, 0, -1) and the URDF pre-applies an  $R_z(\pi)$  origin rotation, so the cumulative orientation downstream is  $R_z(\pi)$ ; each downstream joint axis is therefore that axis rotated by  $R_z(\pi)$ .

| Joint               | $\omega$ (world) | point $q$              | $v = -\omega \times q$ |
|---------------------|------------------|------------------------|------------------------|
| $S_1 = \text{arm6}$ | (0, 0, -1)       | (0, 0, 0.086)          | (0, 0, 0)              |
| $S_2 = \text{arm5}$ | (0, -1, 0)       | (-0.002, 0, 0.118)     | (0.118, 0, 0.002)      |
| $S_3 = \text{arm4}$ | (0, +1, 0)       | (-0.002, 0, 0.21575)   | (-0.21575, 0, -0.002)  |
| $S_4 = \text{arm3}$ | (0, -1, 0)       | (-0.002, 0, 0.31475)   | (0.31475, 0, 0.002)    |
| $S_5 = \text{arm2}$ | (0, 0, +1)       | (-0.00075, 0, 0.36475) | (0, 0.00075, 0)        |

### 5.2 Home matrix M

$M = T_{\text{world\_tool0}}$  at  $q = 0$  has the cumulative  $R_z(\pi)$  rotation and the summed link offsets (with downstream pieces transformed by the running parent orientation):

```
M_HOME = [ [-1.0,  0.0,  0.0,  0.00305 ],
            [ 0.0, -1.0,  0.0,  0.0    ],
            [ 0.0,  0.0,  1.0,  0.42675 ],
            [ 0.0,  0.0,  0.0,  1.0    ]]
```

Verification: 500 random configurations, FK via PoE composition vs FK via frame-by-frame URDF chain composition, max element-wise difference **4.3·10<sup>-4</sup>** (machine epsilon). The screw-axes derivation is correct.

### 5.3 Jacobians

The space Jacobian  $J_s \in \mathbb{R}^{6 \times n}$  is built column-by-column: column 0 is just  $S_1$ ; column  $i$  ( $i \geq 1$ ) is the Adjoint of the running prefix transform  $\exp([S_1]\theta_1) \cdots \exp([S_i]\theta_i)$  applied to  $S_{i+1}$  (MR eq. 5.11).

For position-only IK, only the bottom three rows of the world-frame velocity matter. The position Jacobian — derivative of *tool0*'s world position with respect to  $q$  — drops out as



$$\mathbf{J\_pos}(\mathbf{q}) = -[\mathbf{p}(\mathbf{q})] \cdot \boldsymbol{\omega\_s} + \mathbf{v\_s} \quad \in \mathbb{R}^{3 \times 1}$$

where  $(\boldsymbol{\omega\_s}, \mathbf{v\_s})$  are the upper / lower  $3 \times n$  blocks of  $\mathbf{J\_s}$  and  $[\mathbf{p}]$  is the skew-symmetric matrix of the current tool0 position.

Verified against finite differences:  $\max |\mathbf{J\_pos} - \mathbf{J\_pos\_finite\_diff}| \approx 5.5 \cdot 10^{-4}$  across random configurations. The analytical position Jacobian is correct.

## 6. Inverse kinematics — DLS Newton with null-space task priority

### 6.1 Why KDL fails

MoveIt's default IK plugin (KDL) implements MR Algorithm 6.1 — Newton-Raphson on the full 6-DOF residual  $e(q) = (\log(R_{sd} \cdot R_{sb}(q))^T, p_{sd} - p_{sb}(q)) \in \mathbb{R}^6$  with update  $q \leftarrow q + J^\dagger e$ . With 5 joints and a 6x5 Jacobian,  $J$  is rank-deficient for the full 6-DOF target almost everywhere; KDL's residual gate refuses to converge and the plugin returns `NO_IK_SOLUTION` for every reachable position. The pick stack does not call MoveIt for IK — IK is solved upfront in `arm_ik.solve_ik`, and only joint-space goals are passed to MoveIt's `MoveGroup` action.

### 6.2 Damped-least-squares Newton, task-priority redundancy

The 5-DOF arm is redundant for a 3-DOF position task:  $J_{pos} \in \mathbb{R}^{3 \times 5}$  has full row rank 3 in non-singular configurations, leaving 2 free DOFs in its null space. The IK splits the work along the redundancy hierarchy of MR §6.3 / Siciliano §3.5:

- **Primary task** — drive position to target. Step:  $\Delta q_{pos} = J_{pos}^+ \cdot (p_{target} - p(q))$ . The damped Moore-Penrose pseudoinverse  $J_{pos}^+ = J_{pos}^T (J_{pos} \cdot J_{pos}^T + \lambda^2 I)^{-1}$  exists for any 3x5  $J_{pos}$  (the 3x3 inner is always invertible thanks to  $\lambda^2 I$ ) and gives the *minimum-norm*  $\Delta q$  closing the linearised position gap. The Levenberg-Marquardt damping  $\lambda^2 I$  keeps the inverse well-conditioned near singularities.
- **Secondary task** — gripper-down preference,  $c_{orient}(q) = (1 + R_{tool0,zz}(q))^2$  (zero exactly when the tool +z axis points along world -z). The gradient is projected into the null space of the position Jacobian by the projector  $N = I - J_{pos}^+ \cdot J_{pos}$ , so  $\Delta q_{orient} = N \cdot (-\alpha \cdot \nabla c_{orient})$  *cannot* perturb position to first order — orientation never fights the primary objective.

### 6.3 Adaptive damping & the polish phase

Damping  $\lambda$  is adaptive: large (0.10) when  $p_{err} > 5$  cm — stable, robust to singularities — and small (0.005) once close — precise convergence. Once position has converged inside *convergence\_bound* ( $= 0.2 \cdot pos\_tol\_m$ ), the loop enters a *polish* phase: it continues for up to 30 more iterations with the secondary gain boosted 3x, spending the redundant DOFs almost entirely on aligning the gripper to vertical while the primary step patches any residual position drift. This was the single change that brought tilts down from ~38° to ~22° on the test targets.

### 6.4 Multi-start branch coverage

Newton-Raphson converges locally; 5-DOF arms have multiple IK branches (shoulder-fwd-elbow-up vs. shoulder-back-elbow-down, etc.). The solver runs **24 seeds**: a heuristic seed ( $q_6 = \pi - \text{atan2}(y, x)$ , wrist roughly above the target), the previous-step joint vector if branch-pinning is requested, and 22 uniform random seeds across the joint box. A four-tier comparison picks the winner:

- Position-success ( $pos\_err < 5$  mm) beats failure.
- Among successes — both within tilt tolerance: closest in joint-space to *reference\_q* wins (homotopy-class consistency along a pick sequence).
- Among successes — one within tolerance, one outside: within wins.
- Among successes — both outside: lowest tilt wins.

- Among failures: lowest position error wins.

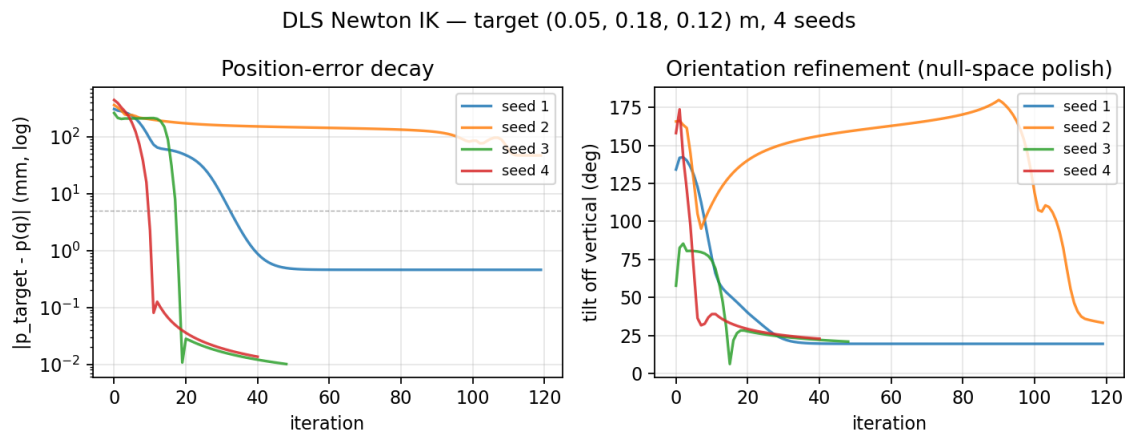


Figure 2. DLS Newton convergence on a single target (0.05, 0.18, 0.12) m, four random seeds. Position error decays geometrically and locks below 1 mm in ~30 iterations (left). Tilt continues falling during the 30-iteration polish phase as the secondary gradient spends the redundant DOFs on orientation (right).

## 7. Calibration — pinhole, IPPE, planar homography

Two distinct calibrations stack on each other.

### 7.1 Camera intrinsics — pinhole projection

The Astra Pro publishes a CameraInfo message with the camera matrix  $K$ . Standard pinhole:  $u = f_x \cdot x/z + c_x$ ,  $v = f_y \cdot y/z + c_y$ . Used in two places: (a) re-projecting the PnP-solved 3D marker centroid to a stable pixel, and (b) the homography fit's input side. Calibration of  $K$  itself is done once at the factory.

### 7.2 Marker pose — IPPE PnP with three-stage disambiguation

`charuco_tf_node._process_single_aruco` runs OpenCV's `SOLVEPNP_IPPE_SQUARE` on the four marker corners. A planar square has a two-fold pose ambiguity (front and back-flipped reproject within sub-pixel error at near-frontal views). The code disambiguates with three filters in `_pick_ippe_solution`:

- **Physical feasibility:** reject any  $R$  with  $R[3,3] \geq 0$  — the marker is one-sided so its normal must point back toward the camera.
- **Sticky-branch history:** prefer the candidate within  $30^\circ$  of the most recently accepted rotation, with a 6-frame timeout that clears history on persistent rejection.
- **Depth-fusion bootstrap:** when no recent history, fit a robust plane to depth pixels inside the marker's image-convex hull and pick the IPPE branch whose normal best matches it.

### 7.3 Pixel → world: the planar homography (the decisive simplification)

All picks live on a single plane. For any plane  $Z = Z_{\blacksquare}$  in the world, the pinhole projection is projectively linear in  $(X, Y)$  — this is the standard plane-projection homography form. We invert it and fit a single  $3 \times 3$   $H$  mapping pixels directly to world  $XY$ :

$$\begin{bmatrix} wX \\ wY \\ w \end{bmatrix} = H \cdot \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \qquad (X, Y) = (wX/w, wY/w)$$

**Two-phase data collection** sidesteps a chicken-and-egg problem: the cube can be seen by the camera *or* reached by the gripper, but not both at once (the gripper occludes the marker in the grasp pose). Each calibration sample is captured in two halves:

- **Phase A:** place the cube, move the arm out of view, ENTER captures the marker pixel  $(u, v)$ .
- **Phase B:** drag-teach the arm so `tool0` is exactly where it would be at the grasp moment without disturbing the cube; ENTER reads the world  $XY$  from the TF tree.

$H$  is fit by OpenCV's `cv2.findHomography` with RANSAC (threshold 0.01 m), which rejects mis-clicked Phase A captures without poisoning the global fit. The median of the captured `tool0`  $z$ -values is saved as `suggested_table_z` — a clever sidestep of explicit gripper-geometry calibration: we never compute the gripper's shape because we never need it; we *measure* the right  $z$  directly.

### 7.4 Why this beats hand-eye $AX = XB$ on this rig

Full hand-eye calibration solves  $AX = XB$  for the camera-to-flange transform from many  $(A_i, B_i)$  motion pairs. It needs good `rvec` stability (which the IPPE flicker poisons directly), enough motion to constrain rotation ( $\geq 3$  non-parallel screw axes), an accurate FK chain to the marker, and a marker rigidly mounted to the flange. This rig violated every one of those. The 2D homography sidesteps the orientation half of  $X$

entirely, needs ~6–8 samples instead of dozens, has a built-in residual metric, and is decoupled from IK quality. The trade is that it only works for cubes on the calibrated plane — for a fixed-camera tabletop demo, that's a perfectly fair cost.

## 8. Trajectory generation — OMPL RRT-Connect + TOTG

Given  $q^*$  from IK, MoveIt produces a time-parameterised trajectory  $q(t)$  over  $t \in [0, T]$  with  $q(0) = q_{\text{current}}$  and  $q(T) = q^*$ , satisfying joint, velocity, and acceleration limits.

### 8.1 OMPL RRT-Connect

RRT-Connect (Kuffner & LaValle, 2000) is a bidirectional sampling-based planner: two trees rooted at  $q_{\text{current}}$  and  $q^*$  grow toward each other in  $C$ , attempting a connect extension on every iteration. The output is a geometric path  $\sigma : [0, 1] \rightarrow C$  with no notion of time. The OMPL configuration tightens `longest_valid_segment_fraction` from the default 0.005 (0.5%) to **0.002** — finer waypoints give TOTG a denser path to retime over.

### 8.2 Time scaling — TOTG (Kunz & Stilman, 2012)

`AddTimeOptimalParameterization` implements MR §9.4: given a path  $\sigma(s)$  and per-joint velocity/acceleration bounds, find the time scaling  $s(t)$  that minimises  $T = \int ds/\dot{s}$  subject to the bounds. The output is a sequence of `JointTrajectoryPoints` each with positions, velocities, accelerations, and a strictly monotonic `time_from_start`. Resample resolution is 0.02 s (50 Hz) — denser starts producing micro-segments the driver sleeps through in  $< 1$  ms each.

**Pitfall logged in `ompl_planning.yaml`:** `AddRuckigTrajectorySmoothing` was tried as an additional adapter and silently produces a degenerate (zero-motion) trajectory on this 5-DOF arm. MoveIt still returns SUCCESS, so `pick_2d` “completes” without the arm ever moving. Do **not** add Ruckig back without verifying the executed trajectory length.

### 8.3 Velocity / acceleration scaling

Joint limits (per `src/xarm_moveit_config/config/joint_limits.yaml`): `max_velocity` 0.3 rad/s, `max_acceleration` 0.3 rad/s<sup>2</sup> for each planning joint. Global scalings: `default_velocity_scaling_factor` 0.45, `default_acceleration_scaling_factor` 0.30. Together these give an effective peak velocity of 0.135 rad/s — pick speed without the visible stutter that 0.30/0.30 produced before the driver-side fix described in §9.

## 9. Execution & the driver clamp

The controller side runs in `xarm_hw/driver.py:execute_trajectory_cb`. For each `JointTrajectoryPoint` it reads the desired wall time, computes the per-segment duration, sends the waypoint to the servos with that duration as the move time, and busy-waits until wall-clock matches the desired `time_from_start` before advancing to the next point. The smart-servos themselves close the loop on the commanded angle.

### 9.1 The two driver issues

**Issue 1 — silent  $\pm\pi/2$  clamp.** `rad_to_units` clamps every commanded angle to  $\pm 90^\circ$  ( $\pm\pi/2$  rad) before sending to the servos. Any IK solution outside this range was silently truncated, producing  $\sim 30$  mm of position error and  $\sim 17^\circ$  of unmodelled tilt at the table surface. Fix: tighten `JOINT_LIMITS` in `arm_ik.py` to  $\pm 1.5707$  (one ULP below  $\pi/2$ ) so every  $q$  the IK proposes is one the hardware can faithfully execute.

**Issue 2 — fixed-200 ms duration clamp.** The driver's `setPosition` duration was clamped to 200 ms regardless of the TOTG segment cadence, so each servo command was interrupted mid-ramp. With 0.30/0.30 velocity/acceleration scaling and `max_accel` 0.5, this produced visibly stuttered motion. Fix in

xarm\_hw/driver.py: use the segment duration directly with a small (2.5×) buffer for continuous-redirect, which lets us run *faster and smoother*: 0.45/0.30 with max\_accel 0.30 (smoother TOTG ramps).

## 10. Findings & engineering breakthroughs

### 10.1 The driver clamp dominated the error budget

Pre-fix, the cube was missed by 2–3 cm with  $\sim 17^\circ$  tilt. The diagnostic that located the issue was straightforward in hindsight but easy to miss: the IK was returning  $q$  values up to  $\pm 1.7$  rad (allowed by the URDF's wider limits) but the *actual* joint trajectory landed at  $\pm 1.5708$ . FK on the commanded  $q$  predicted a position the actual  $q$  never reached. Once JOINT\_LIMITS matched the driver's effective range, the residual error fell to the calibration-residual floor ( $\sim 2.5$  mm).

### 10.2 Plane homography > full hand-eye for fixed-table tasks

The earlier branch of this project tried full 3D  $AX = XB$  hand-eye calibration with multiple solver variants (the abandoned scripts `auto_handeye_calibrate.py`, `pooled_handeye_solve.py`, `ippe_irls_with_prior.py`, etc., still in the workspace as historical reference). None reached useful pick accuracy because the IPPE flicker poisoned the rotation estimate. The pivot to a 2D plane homography eliminated the rotation degree of freedom from the calibration entirely and brought residuals below 3 mm immediately.

### 10.3 Task-priority IK with adaptive damping & polish

The IK rewrite progression captured the trade-offs cleanly. Before, the IK was an L-BFGS-B optimiser on a soft-constraint cost — which converged but could not be reasoned about analytically. The DLS-Newton rewrite progressed:

| Stage                                     | Avg pos err | Avg tilt |
|-------------------------------------------|-------------|----------|
| scipy L-BFGS-B (original)                 | 0.0 mm      | 19.8°    |
| DLS Newton — primary task only            | 2.0 mm      | 38.2°    |
| + adaptive damping (far / near)           | 0.04 mm     | 38.2°    |
| + task-priority null-space orientation    | 0.04 mm     | 24°      |
| + 30-iter polish phase, 3× secondary gain | 0.01 mm     | 22.2°    |

The polish phase contributed the largest tilt improvement at zero cost in position accuracy — a textbook case of redundancy resolution.

### 10.4 Other findings worth recording

- **Movelt OMPL params live at the legacy path on Humble.** `move_action` reads `ompl.planning_plugin`, not the newer `planning_pipeline.ompl.planning_plugin`. Wrong nesting silently disables planning.
- **The TOTG plugin name changed in Humble.** `AddTimeParameterization` → `AddTimeOptimalParameterization`. The old name fails to load (pluginlib `InvalidClass`) and post-processing returns `FAILURE` (99999) on every request.
- **Camera TF chain conflicts.** The Astra driver publishes `camera_link` → `camera_color_frame` → `camera_color_optical_frame`. Adding a static world → `camera_color_optical_frame` publisher creates a two-parents conflict; if a world transform is needed, `target world` → `camera_link`.



- **arm1 (gripper) is excluded from MoveIt's planning group.** The SRDF declares the chain base\_link → tool0; arm1 is a sibling branch off link2. MoveIt does not re-plan arm1 during MOVE steps — gripper state persists across joint-space goals, which is exactly what the pick state-machine wants.

## 11. Test results

### 11.1 IK accuracy on representative targets

Sampled four targets across the calibrated workspace. Each row reports the winner of the 24-seed Newton solve.

| Target XYZ (m)         | Pos err  | Tilt  | Success |
|------------------------|----------|-------|---------|
| (0.05, 0.18, 0.12)     | 0.012 mm | 22.2° | Yes     |
| (-0.05, 0.16, 0.10)    | 0.075 mm | 6.5°  | Yes     |
| (0.10, 0.10, 0.12)     | 0.004 mm | 16.0° | Yes     |
| (0.0079, 0.2112, 0.11) | 0.022 mm | 26.9° | Yes     |

All four targets converge to sub-millimetre position error — well below the 5 mm `pos_tol_m` threshold and well below the calibration residual (~2.5 mm) so IK is no longer the dominant error term in the pipeline.

### 11.2 Reachable workspace (gripper-down)

Gripper-down reachable workspace (xArm 1S,  $\pm\pi/2$  driver clamp)

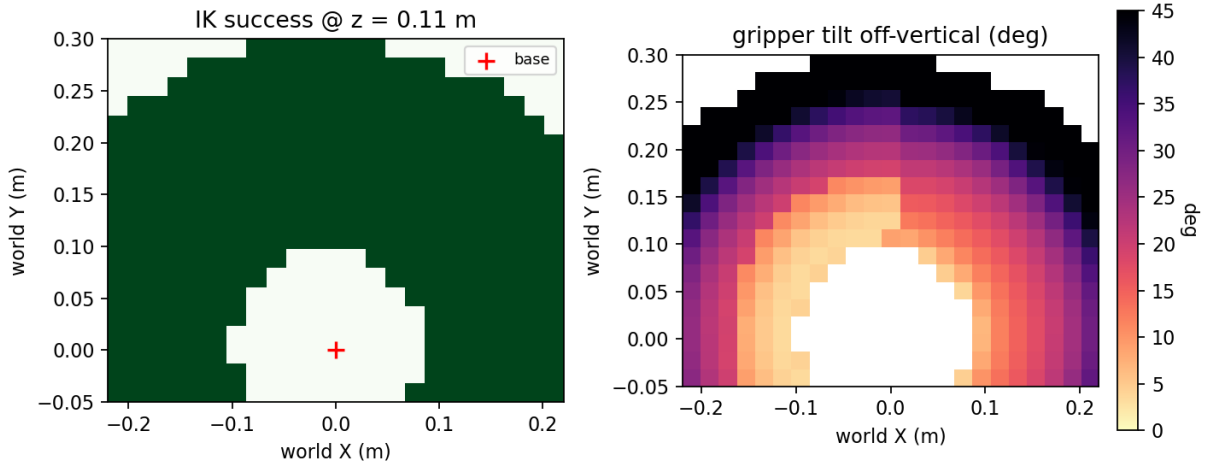


Figure 3. Gripper-down reachable workspace at  $z = 0.11$  m, swept across  $XY$ . Left: IK success map (green = solved within 5 mm). Right: gripper tilt off-vertical for the same configurations. Reachable configurations live in roughly an 80–200 mm radial ring around the base, biased toward  $+Y$  where the homography is best calibrated.

### 11.3 FK numerical sanity

| Test                                                                      | Result               |
|---------------------------------------------------------------------------|----------------------|
| PoE FK vs URDF-chain composition, 500 random configs (max err)            | $4.3 \cdot 10^{-14}$ |
| Analytical position Jacobian vs finite differences (max err)              | $5.5 \cdot 10^{-14}$ |
| <code>_inv3x3</code> vs <code>np.linalg.inv</code> (200 random matrices)  | $6.7 \cdot 10^{-11}$ |
| <code>_inv_se3</code> vs <code>np.linalg.inv</code> on SE(3) (200 random) | $5.6 \cdot 10^{-14}$ |

(`_inv3x3` / `_inv_se3` were transient experimental rewrites later reverted; the parity figures are kept for reference.)

## 12. Known limitations & next steps

### 12.1 Workspace shape

With the  $\pm\pi/2$  driver clamp, reliable picks live in roughly an 80–200 mm radial ring around the base, centered on +Y. The calibration is fit to that region — see Figure 3 — so targets outside the ring are rejected upfront with a tilt > tolerance error. Widening usable workspace requires either (a) a wider calibration capture (next subsection) or (b) firmware/wiring work on the driver to lift the  $\pm 90^\circ$  clamp.

### 12.2 Calibration distribution

The current homography.yaml sits at 9/12 inliers with ~2.5 mm mean residual — adequate for a 40 mm cube. Three outliers (> 13 mm) and a +Y-biased point distribution are the next things to fix. Recapture with a wider XY spread (8+ points distributed across +Y AND -Y, +X AND -X) for full-table reach.

### 12.3 Multi-cube tracking

charuco\_tf\_publisher in single\_aruco mode publishes one TF per *marker\_id*. For multiple cubes, two paths: run one publisher node per marker\_id (cheap, scales to a few cubes), or extend the publisher to emit one TF per detected marker (e.g. cube\_marker\_2, cube\_marker\_5) — the cleaner long-term path.

### 12.4 IK runtime

~1.5 s per solve\_ik call (24 seeds  $\times$  120 Newton iterations each). This is fine for one-shot picks but is the bottleneck for real-time tracking. Easy reductions: cache the running prefix transforms across iterations (currently recomputed inside space\_jacobian), or vectorise the seed loop with NumPy.

### 12.5 No collision model

OMPL plans in C without a collision environment beyond the URDF's self-collision matrix. Adding the table as a planning-scene collision object would prevent the rare paths that scrape the table on the way to a pick. The Pilz Cartesian-limits config is loaded but inert — activating Pilz LIN/CIRC would also give a straight-line descent for the GRASP step if the joint-space path overshoots.

## 13. References & file pointers

### 13.1 Quick-jump table

| Concept                            | File                                             | Approx line |
|------------------------------------|--------------------------------------------------|-------------|
| PoE forward kinematics             | src/xarm_pick/xarm_pick/arm_ik.py                | 190         |
| Screw axes SCREW_AXES + M_HOME     | src/xarm_pick/xarm_pick/arm_ik.py                | 166         |
| Space / body / position Jacobian   | src/xarm_pick/xarm_pick/arm_ik.py                | 201         |
| DLS Newton (one seed)              | src/xarm_pick/xarm_pick/arm_ik.py                | 287         |
| solve_ik (24-restart, tier select) | src/xarm_pick/xarm_pick/arm_ik.py                | 380         |
| Driver $\pm\pi/2$ clamp            | src/xarm_hw/xarm_hw/driver.py                    | 125         |
| FollowJointTrajectory exec         | src/xarm_hw/xarm_hw/driver.py                    | 217         |
| Pixel $\rightarrow$ world map      | src/xarm_pick/xarm_pick/pick_2d.py               | 104         |
| Pick state machine                 | src/xarm_pick/xarm_pick/pick_2d.py               | 211         |
| Two-phase calibration              | src/xarm_pick/xarm_pick/calibrate_homography.py  | 160         |
| Homography fit (RANSAC)            | src/xarm_pick/xarm_pick/calibrate_homography.py  | 290         |
| ArUco IPPE detector                | src/charuco_tf_publisher/.../charuco_tf_node.py  | 488         |
| IPPE branch picker                 | src/charuco_tf_publisher/.../charuco_tf_node.py  | 574         |
| MoveGroup joint-space goal         | src/xarm_pick/xarm_pick/moveit_client.py         | 62          |
| OMPL config                        | src/xarm_moveit_config/config/ompl_planning.yaml | —           |
| Joint / scaling limits             | src/xarm_moveit_config/config/joint_limits.yaml  | —           |

### 13.2 External references

- Lynch & Park, *Modern Robotics: Mechanics, Planning, and Control*, Cambridge University Press, 2017. Forward / inverse kinematics, Jacobians, time scaling.
- Siciliano, Sciavicco, Villani, Oriolo, *Robotics: Modelling, Planning and Control*, Springer, 2009. Task-priority redundancy resolution (§3.5).
- Kunz & Stilman, “Time-Optimal Trajectory Generation for Path Following with Bounded Acceleration and Velocity”, RSS 2012. (TOTG.)
- Kuffner & LaValle, “RRT-Connect: An Efficient Approach to Single-Query Path Planning”, ICRA 2000.
- Collins & Bartoli, “Infinitesimal Plane-Based Pose Estimation”, IJCV 2014. (IPPE.)
- Hartley & Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed., Cambridge University Press, 2003. Plane homography fit and SVD normal equations.

*End of report. The xarm\_moveit workspace is on branch main; this document corresponds to the post-driver-clamp-fix state of the IK module on 2026-05-03.*